was the controlling factor to carry the characteristics from one generation to another.

Genetics, a branch of science aims to understand the working of the controlling factor. The factor governing the transfer of information is now very well known as Gene. Gene can be defined as a unit of heredity which is transferred from parent to progeny.

Although there were experiments to prove the inheritance pattern due to gene, there was no real understanding of the molecular nature of the gene. Work of Frederick Griffith introduced the transforming principle which was further confirmed as Deoxyribonucleic acid (DNA) by experiments of Avery, Mac Leod and Mac Carty in 1944 followed by Hershey and Chase in 1952.

14.1.1 Frederick Griffith's Experiment

In 1928, British bacteriologist Frederick Griffith (Figure 14.1) was trying to develop a vaccine against pneumonia. In his experiments Griffith used two related strains of *Streptococcus pneumoniae* (Figure 14.2).



Figure 14.1: Frederick Griffith

1. Rough strain (R) – avirulent, non-capsulated strain, forming rough colonies on culture media.
2. Smooth strain (S) – virulent, capsulated strain (resists phagocytosis), forming smooth colonies on culture media.

Griffith injected live smooth strain into mice which caused disease and killed it. When he injected live rough strain into mice, it did not cause disease and mice remained alive. He heat killed the smooth strain and injected into mice, the mice remained alive. But the experiment gave surprising results when a mixture of harmless live rough strain and heat-killed smooth strain was injected into mice. Griffith observed that the mice developed pneumonia and died. Further, when he analysed the blood sample from dead mouse, he found that it contained live smooth strain. This accidental discovery made Griffith to conclude that the rough strain changed (transformed) into smooth strain by taking up a substance which he called a "transforming principle" from heat killed smooth bacteria. This phenomenon is called "Bacterial Transformation" Griffith's experiment is summarized in Figure 14.3.

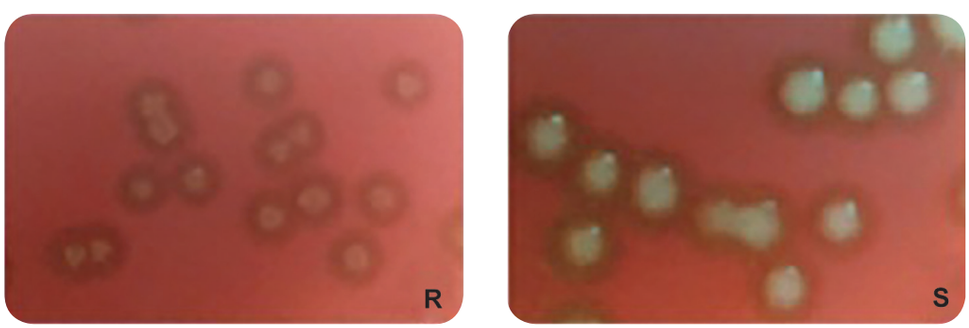


Figure 14.2: Rough and Smooth colonies of *Streptococcus pneumoniae*

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